VAI PRICING AGENT

Pricing decision support for portfolio review, SKU diagnostics, and simulation.

The system combines workbook-based retail pricing data, business rules, simulation logic, and an analyst-facing web interface to help teams review pricing opportunities, understand recommendation drivers, and test changes before execution.

WHAT THE SYSTEM DOES

1. Loads pricing workbooks and normalizes SKU data

Excel sheets are parsed into a consistent record structure including price, cost, margin floor, competitor prices, KVI flag, inventory, and source workbook signals.

2. Generates recommendations at SKU level

The engine recommends increase, decrease, or hold by balancing market position, margin protection, demand sensitivity, and inventory context.

3. Simulates commercial outcomes

Users can see projected volume change, expected units sold, revenue impact, margin impact, and projected margin at any proposed price.

4. Supports analyst review and action

The frontend provides dashboard views, SKU detail pages, agent review prioritization, and an SKU Q&A interface for explanation and follow-up.

PRIMARY OUTCOMES

- Faster identification of margin risks and overpriced / underpriced SKUs
- Consistent recommendation logic across the pricing queue
- Transparent explanation of why a recommendation was made
- Ability to test alternate prices before commercial sign-off
- Portfolio-level visibility combined with SKU-level detail

USER WORKFLOW

01
Import dataset
Workbook sheets such as NEW, NEW (2), and Pricing Engine V1 are read and normalized.

02
Generate recommendation
The engine selects the best action and suggested price for each SKU.

03
Explain the drivers
Users inspect margin, competitor pricing, stock cover, and source workbook context.

04
Run simulation
Alternative prices can be tested to compare demand and profitability outcomes.

05
Review and act
Teams use the queue, flagged SKUs, and agent review panel to prioritize execution.

DECISION TYPES

CORE SIMULATION METRICS

MAIN DATA SIGNALS

REVIEW SURFACES

3
Increase, decrease, hold

5
Volume, units, revenue, margin $, margin %

10+
Price, cost, market, flags, inventory, dates

4
Dashboard, queue, SKU detail, agent review

This summary is intended for stakeholder review and solution overview. Page 2 contains technical details on architecture, simulation logic, and current implementation choices.

TECHNICAL DETAIL

How the system is built and how the recommendation logic works.

The current implementation is a local full-stack application with a FastAPI backend, a React frontend, workbook parsing logic, a recommendation engine, and a simulation module that estimates the effect of price changes.

ARCHITECTURE

Frontend	React + Vite interface for dashboard, SKU queue, SKU detail, chat, and manual simulation.
Backend API	FastAPI service that loads the active dataset, generates recommendations, and serves dashboard / SKU endpoints.
Data loader	Workbook parsing logic that extracts supported sheets and maps raw columns into typed SKU records.
Decision engine	Candidate-price evaluation using margin, market gap, revenue impact, margin-dollar impact, and business guardrails.
Simulation engine	Rule-based elasticity model that estimates unit change and commercial outcomes at a proposed price.
Vector / LLM layer	Local vector store for recommendation context plus optional LLM support for SKU-level question answering.

CURRENT LOGIC HIGHLIGHTS

- Demand elasticity is estimated from KVI status, promo status, stock cover, market gap, and source pricing flags.
- The engine scores multiple candidate prices rather than applying a single fixed rule.
- Recommendations are constrained by margin floor, market distance, and sensible move size.
- Simulation outputs include rounded-up expected units sold and projected gross margin percentage.
- Workbook-driven fields such as stock cover and inventory interpretation are now read from updated sheet formats.

RECOMMENDATION AND SIMULATION FORMULA SUMMARY

EXPECTED VOLUME CHANGE

Price change % × estimated demand elasticity

EXPECTED UNITS SOLD

Last week units × (1 + expected volume change), rounded up

EXPECTED REVENUE IMPACT

Projected revenue − current revenue

EXPECTED MARGIN IMPACT

Projected gross profit − current gross profit

PROJECTED MARGIN

(Proposed price − cost) / proposed price

RECOMMENDATION CHOICE

Best-scoring candidate after revenue, margin, market, and guardrail checks

OPERATIONAL BENEFITS

- Analysts can explain recommendations with evidence rather than static rules alone.
- Pricing decisions are linked to commercial impact, not just competitor gap.
- The interface helps teams move from portfolio review to SKU action without changing tools.
- Workbook refreshes can be incorporated quickly through supported sheet mappings.

CURRENT SCOPE AND NEXT ENHANCEMENTS

- Rule-based elasticity can later be calibrated with historical price / sales data.
- Confidence scoring can be refined using backtesting accuracy.
- Executive reporting can be expanded with exportable summaries and approval workflows.
- Recommendation governance can include category-specific rules and commercial override logging.

Prepared as a concise two-page overview of the current VAI Pricing Agent implementation. Source file is editable HTML so the brief can be revised or re-exported as the system evolves.